Guangzhou Food Guide - Automated Content Management System

1. Theme and Problems

- **Theme:** Build an automated content management system that uses an SQLite database to store and manage Guangzhou food establishment data, and generates food recommendation reports based on predefined templates. These reports will showcase key information such as restaurant rankings, ratings, and recommended dishes across different business districts and cuisine types, helping consumers make more informed choices.
- **Core Problems:**
    1. **Data Storage and Management:** How to efficiently and standardizedly store structured food establishment information (name, ratings, price, cuisine type, business district, recommended dishes, etc.) and report templates in an SQLite database?
    2. **Data Query and Analysis:** How to support flexible data queries (basic filtering, aggregate statistics, subqueries) to extract valuable information (e.g., popular business districts, cost-effective establishments, cuisine rating comparisons)?
    3. **Template-Driven Report Generation:** How to design reusable report templates (stored in the database) and use SQL query results (particularly those containing aggregates and subqueries) to dynamically populate these templates, generating reports in plain text, Markdown, or HTML format, or presenting them visually with charts?
    4. **System Integration and Interaction:** How to build a complete Python application to manage data (insertion, querying), manage templates, and execute the report generation workflow?

2. Dataset

- **Source:** This dataset comes from the Heywhale platform and contains detailed information about food establishments in Guangzhou, including restaurant names, number of reviews, prices, ratings, cuisine types, business districts, etc., for food recommendations and analysis.
- **Content:** Contains detailed information for numerous Guangzhou food establishments with fields including:
    - Name (Restaurant Name)
    - Reviews (Number of Reviews)
    - Avg price (RMB) (Average Price per Person - RMB)
    - Category (Cuisine Type - e.g., Cantonese, Japanese, Hot Pot, etc.)
    - Business district (Business District)
    - Address (Address)
    - Recommendation 1/2/3 (Recommended Dishes 1/2/3)
    - Taste score (Taste Rating)

- Environment score (Environment Rating)
- Service score (Service Rating)
- Star level (Star Rating - e.g., Five-Star Merchant, Quasi-Five-Star Merchant)
- Shop ID (Store ID)
- URL (Dianping Link)

3. Templates

1. Template 1: Star Rating Analysis Report

- **Metadata:**
  - `template_id=1`
  - `title='Guangzhou Catering Star Rating Distribution Analysis'`
  - `keywords='star rating,score,industry level,data analysis'`
  - `default_format='markdown'`
- **Template Content (with placeholders):**

```
# Guangzhou Catering Star Rating Distribution Analysis

The Guangzhou food scene is full of hidden gems! Based on data analysis, we provide
an in-depth look at the local catering star rating distribution pattern:

### **Star Rating Distribution Overview**
**Five-Star Merchant Percentage:** {percentage_five_star:.1f}% - Represents top-tier
dining experiences
**Quasi-Five-Star Merchant Percentage:** {percentage_quasi_five:.1f}% - Similarly
excellent quality choices
**Four-Star Merchant Count:** {count_four_star} establishments - Reliable quality
despite smaller numbers

### **Rating Performance by Star Level**
**Five-Star Merchant Average Ratings:**
**Taste:** {avg_taste_five:.1f}/10
**Environment:** {avg_env_five:.1f}/10
**Service:** {avg_service_five:.1f}/10

**Quasi-Five-Star Merchant Average Ratings:**
**Taste:** {avg_taste_quasi:.1f}/10
**Environment:** {avg_env_quasi:.1f}/10
**Service:** {avg_service_quasi:.1f}/10

**Four-Star Merchant Highlight:** Environment and service average score reaches **
{avg_env_service_four_star:.1f}**, showing commendable performance

### **Overall Industry Standards**
Guangzhou catering maintains high standards in these dimensions:
```

```
**Taste:** {avg_taste_overall:.1f}/10
**Environment:** {avg_env_overall:.1f}/10
**Service:** {avg_service_overall:.1f}/10

> *Data Insight:* Analysis based on a sample of {total_restaurants} restaurants shows Five-Star Merchants particularly excel in environment ratings.
```

2. Template 2: Best Value for Money under X Yuan

- **Metadata:**
  - `template_id=2`
  - `title='Best Value Recommendations under {price_cap} Yuan'`
  - `keywords='cost-performance,price ceiling,{price_cap}yuan,taste rating'`
  - `default_format='markdown'`
- **Template Content (with placeholders):**

```
# Best Value Recommendations under {price_cap} Yuan

Based on your budget, we've selected the {top_n} restaurants with the best value for money under {price_cap} yuan! (Value determined by taste rating to price ratio)

**{restaurant1_name}** (Value Index: {value_for_money1:.2f})
**Taste Rating:** {restaurant1_taste_score:.1f}
**Avg Price:** ¥{restaurant1_price}
**Recommended Dishes:** `{restaurant1_rec1}`, `{restaurant1_rec2}`

**{restaurant2_name}** (Value Index: {value_for_money2:.2f})
**Taste Rating:** {restaurant2_taste_score:.1f}
**Avg Price:** ¥{restaurant2_price}
**Recommended Dishes:** `{restaurant2_rec1}`, `{restaurant2_rec2}`

**{restaurant3_name}** (Value Index: {value_for_money3:.2f})
**Taste Rating:** {restaurant3_taste_score:.1f}
**Avg Price:** ¥{restaurant3_price}
**Recommended Dishes:** `{restaurant3_rec1}`, `{restaurant3_rec2}`

**{restaurant4_name}** (Value Index: {value_for_money4:.2f})
**Taste Rating:** {restaurant4_taste_score:.1f}
**Avg Price:** ¥{restaurant4_price}
**Recommended Dishes:** `{restaurant4_rec1}`, `{restaurant4_rec2}`

**{restaurant5_name}** (Value Index: {value_for_money5:.2f})
**Taste Rating:** {restaurant5_taste_score:.1f}
**Avg Price:** ¥{restaurant5_price}
**Recommended Dishes:** `{restaurant5_rec1}`, `{restaurant5_rec2}`

> *Tip:* These restaurants provide the best taste experience on a limited budget - a smart choice!
```

3. Template 3: Winter Warmers

- **Metadata:**
  - `template_id=3`
  - `title='Dispelling the Chill: Winter Warming Food Recommendations in {district} District'`
  - `keywords='winter,warming stomach,hot pot,casserole,{district}'`
  - `default_format='text'`
- **Template Content (with placeholders):**

```
As winter arrives, the {district} district offers a guide to warming foods! These steaming hot choices will warm you from the stomach to the heart:

[Hot Pot First Choice] {hotpot_restaurant_name} ({hotpot_restaurant_score:.1f}) - ¥{hotpot_price}/person
Signature Hot Pot: {hotpot_rec1} (Must-try) & {hotpot_rec2}
Address: {hotpot_address}

[Rich Soup Casseroles] {stew_restaurant_name} ({stew_restaurant_score:.1f}) - ¥{stew_price}/person
Warming Casserole: {stew_rec1} (Rich broth, delicious flavor) & {stew_rec2}
Address: {stew_address}
```

4. Template 4: Hidden Gems

- **Metadata:**
  - `template_id=4`
  - `title='Alleyway Surprises: Discovering Small, High-Rated Restaurants in {district} District'`
  - `keywords='small,high value,treasure stores,{district}'`
  - `default_format='markdown'`
- **Template Content (with placeholders):**

```
# Alleyway Surprises: Discovering Small, High-Rated Restaurants in {district} District

Tired of crowded stores? The {district} district hides some treasure restaurants that are less reviewed but highly rated! They might be less conspicuous but could offer unexpected delights:

**{gem_name}**
**Type:** {gem_category}
**Rating:** **{gem_score:.1f}** (Taste:{gem_taste} Environment:{gem_env} Service: {gem_service})
**Avg Price:** ¥{gem_price} (Great value)
**Highlights:** Owner recommends `{gem_rec1}` and `{gem_rec2}`, reportedly {gem_highlight}.
**Address:** {gem_address}
```

5. Template 5: District Food Express

- **Metadata:**
  - `template_id=5`
  - `title='District Food Express: {district}'`
  - `keywords='overview,popular cuisines,average price,highly rated stores'`
  - `default_format='html'`
- **Template Content (with placeholders):**

```
<h2>District Food Express: {district}</h2>
<p>In the vibrant <strong>{district}</strong> business district, the food map is being refreshed! Data insights show:
The main consumption driver here is <strong>{most_common_category}</strong> cuisine (accounting for 42%),
while also offering diverse choices like <strong>{distinct_category}</strong>.</p>
<p>The overall consumption level is affordable, averaging about <strong>{avg_price_district:.1f}</strong> yuan.
The most notable highly-rated representatives include:</p>
<ul>
  <li><strong>{restaurant1_name}</strong> (Comprehensive Rating: {restaurant1_score:.1f})</li>
  <li><strong>{restaurant2_name}</strong> (Comprehensive Rating: {restaurant2_score:.1f})</li>
</ul>
<p><strong>Dining Tip:</strong> The peak dining hours in this district are concentrated between 18:00-19:30, it's advised to make reservations in advance for popular stores like {restaurant1_name}.</p>
<!-- Optional: Business district heatmap anchor point -->
```

6. Template 6: Top Delivery Picks

- **Metadata:**
  - `template_id=6`
  - `title='Food at Home: Top Rated Delivery Picks in {district} District'`
  - `keywords='delivery,home,convenient,high rating'`
  - `default_format='markdown'`
- **Template Content (with placeholders):**

```
# Food at Home: Top Rated Delivery Picks in {district} District

Enjoy the deliciousness of the **{district}** district without leaving home! We've
selected several restaurants with **high comprehensive ratings ({min_score}+)**
suitable for delivery:

**{restaurant1_name} ({restaurant1_score:.1f})** - {restaurant1_category}
**Avg. Price:** ¥{restaurant1_price}
**Signature Delivery Dishes:** `{restaurant1_rec1}`, `{restaurant1_rec2}`

**{restaurant2_name} ({restaurant2_score:.1f})** - {restaurant2_category}
**Avg. Price:** ¥{restaurant2_price}
**Signature Delivery Dishes:** `{restaurant2_rec1}`, `{restaurant2_rec2}`

**{restaurant3_name} ({restaurant3_score:.1f})** - {restaurant3_category}
**Avg. Price:** ¥{restaurant3_price}
**Signature Delivery Dishes:** `{restaurant3_rec1}`, `{restaurant3_rec2}`

> *Tip:* The above restaurants excel in both **Taste ({avg_taste_score:.1f})** and
**Service ({avg_service_score:.1f})**, making them great choices for delivery!
```

4. Automated Report/News Sketch

Using Template 1 as an example:

```
# Guangzhou Catering Star Rating Distribution Analysis

The Guangzhou food scene is full of hidden gems! Based on data analysis, we provide an in-
depth look at the local catering star rating distribution pattern:

### **Star Rating Distribution Overview**
**Five-Star Merchant Percentage:** 32.3% - Represents top-tier dining experiences
**Quasi-Five-Star Merchant Percentage:** 42.6% - Similarly excellent quality choices
**Four-Star Merchant Count:** 25 establishments - Reliable quality despite smaller numbers

### **Rating Performance by Star Level**
**Five-Star Merchant Average Ratings:**
**Taste:** 8.8/10
**Environment:** 8.9/10
**Service:** 8.9/10
```

Quasi-Five-Star Merchant Average Ratings:
Taste: 8.6/10
Environment: 8.7/10
Service: 8.7/10

Four-Star Merchant Highlight: Environment and service average score reaches **8.3**, showing commendable performance

Overall Industry Standards
Guangzhou catering maintains high standards in these dimensions:
Taste: 8.7/10
Environment: 8.7/10
Service: 8.7/10

> *Data Insight:* Analysis based on a sample of 93 restaurants shows Five-Star Merchants particularly excel in environment ratings.